

Roger Richard Demello

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SUMMARY

AI Engineer with experience **building LLM applications**, RAG systems, and autonomous AI agents. Developed and deployed 3+ production grade AI solutions serving **200+** users using FastAPI, LangChain, vector databases, and cloud technologies. Skilled in **backend** engineering, machine learning, and AI product development, with a focus on **building scalable systems** that solve real-world business problems.

WORK EXPERIENCE

AI LifeBOT

Jan 2026 - Present

AI Engineer Intern

- Built and **deployed 3+** production-grade LLM applications and autonomous AI agents, serving **200+** users and improving task completion **efficiency** through tool integration, long-term memory, and goal-driven **workflows**.
- Architected end-to-end **AI pipelines** integrating LLMs, RAG, external APIs, and streaming layers, reducing average response **latency** by **35%** and improving **system reliability**.
- Led rapid prototyping of **5+** AI features, collaborating with cross-functional teams to **accelerate** product delivery and **validation** cycles.

CFM, RCOEM

May 2025 - Jul 2025

Machine Learning Intern

- Improved a sleep disorder **prediction** system using **1,000+** health records and feature engineering techniques, increasing model robustness and generalization.
- Achieved **87% prediction accuracy** through structured feature engineering and statistical analysis of stress levels, sleep duration, and blood pressure.
- Built, trained, and **evaluated** models in Python with Scikit-learn and Pandas, optimizing end-to-end data workflows to cut development time by **30%** and increase model **validation** accuracy by **25%**, leading to faster, more reliable insights.

PROJECTS

Autonomous Executive Email Copilot | [GitHub](#)

- Engineered a **deterministic RL-style executive inbox** simulation environment for **benchmarking** autonomous email agents across classification, **prioritization**, and action workflows.
- Developed multi-agent orchestration with **FastAPI**, **vector retrieval**, approval gates, and **telemetry pipelines** for reproducible evaluation and performance monitoring.
- Built policy-driven email management workflows achieving context-aware prioritization through RAG and tool-integrated decision making.

DealSentry - Intelligent Proposal Guard | [GitHub](#)

- Built an enterprise AI-powered **compliance and risk analysis platform** used to automate proposal validation and **reduce** manual review effort using React, TypeScript, Node.js, PostgreSQL, and Azure OpenAI.
- Designed a rule-based compliance engine with AI-driven risk scoring, **document ingestion** (DOCX/PDF), approval workflows, audit trails, and contract lifecycle tracking.
- Integrated HubSpot and Gmail APIs, reducing manual compliance review effort and accelerating approval workflows.

SentinelOps - Autonomous Incident Prediction & Prevention Platform | [GitHub](#)

- Built an AI-powered platform using FastAPI, React, and **anomaly detection** models to analyze **10,000+** **synthetic** telemetry events and **automate** root-cause analysis.
- Engineered multi-agent workflows for anomaly detection, impact assessment, and remediation **recommendations**.
- Integrated **observability** pipelines, **audit** logging, and Slack/PagerDuty alerts for proactive **infrastructure monitoring**.

SKILLS

- Agentic & Generative AI:** RAG Pipelines, Vector Databases, Multi-Agent Systems, Prompt Engineering, Agentic Workflows
- Machine Learning & AI:** Scikit-learn, NumPy, Pandas, PyTorch, LLMs, Semantic Search
- Backend & DevOps:** Amazon Web Services, FastAPI, Flask, REST APIs, CI/CD, PostgreSQL, Vercel
- Programming Languages:** C++, Python, Java, C, SQL
- Tools & Platforms:** Git, GitHub, Hugging Face, Supabase, LangChain, Docker

EDUCATION

Shri Ramdeobaba College of Engineering and Management

Nov 2022 - May 2026

Bachelor of Technology, Electronics and Communication

- GPA:** 8.90/10

Shri Ramdeobaba College of Engineering and Management

Jun 2023 - Dec 2025

Minor Degree: Artificial Intelligence and Machine Learning

- GPA:** 9.67/10

CERTIFICATIONS / AWARDS

- AWS Certified Cloud Practitioner:** Amazon Web Services (Issued October 2025)
- Finalist: **Paytm x Sarvam x Logitech** AI National Hackathon
- 2nd Place: **Bytesage** AI National Hackathon